

PROJECT TITLE

Diabetes Prediction System

ABSTRACT

Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels, which, if left untreated, can lead to serious health complications such as kidney failure, cardiovascular diseases, nerve damage, and vision problems. Early detection and timely intervention are crucial to prevent these adverse effects and improve the quality of life for patients. Traditional diagnostic methods rely on laboratory tests and clinical assessments, which are often time-consuming, expensive, and not easily accessible, particularly in rural or remote areas. Manual interpretation of medical data may also result in errors or delayed diagnosis, affecting patient care. The **Diabetes Prediction System** is a machine learning-based solution developed using Python that predicts the likelihood of an individual having diabetes based on health parameters such as glucose levels, body mass index (BMI), blood pressure, insulin levels, age, and family history. By analyzing historical patient datasets, the system trains supervised learning algorithms to recognize patterns and correlations, enabling accurate predictions for new patients. This automated approach reduces manual effort, minimizes errors, and provides immediate insights into a patient's risk. The system can also incorporate data visualization and a web-based interface, making it accessible and user-friendly. Overall, this project demonstrates the practical application of artificial intelligence in healthcare, facilitating early diagnosis, better patient management, and preventive care.

PROBLEM STATEMENT

Diabetes is a major health concern worldwide, affecting millions of individuals and causing severe complications if not diagnosed early. Conventional diagnostic methods such as laboratory tests and clinical evaluations can be time-consuming, costly, and inaccessible, especially in underprivileged areas. Manual analysis of patient data may also result in errors or delayed diagnosis, limiting the effectiveness of preventive measures. There is a need for an automated system that can predict the risk of diabetes using measurable health parameters such as glucose levels, BMI, blood pressure, insulin levels, age, and family history. Such a

system should provide accurate, reliable, and timely predictions, enabling healthcare professionals and patients to take preventive or therapeutic actions, reduce complications, and improve overall patient outcomes.

SYSTEM REQUIREMENTS

Software Requirements

- Operating System: Windows 11
- Programming Language: Python
- Python Libraries:
 - pandas (for data handling)
 - numpy (for numerical operations)
 - scikit-learn (for machine learning algorithms)
 - matplotlib / seaborn (for data visualization)
 - Flask / Django (optional, for web interface)
- IDE / Editor: VS Code
- Database / Storage: CSV file or simple database to store patient records

Hardware Requirements

- Processor: Minimum Intel i3 (8th generation or higher recommended)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: Minimum 10 GB free disk space
- Internet: Optional (for dataset download or web interface)

Functional Requirements

- Input patient health parameters such as glucose, BMI, blood pressure, insulin, and age
- Predict diabetes status (Positive / Negative)

- Visualize trends in patient data (optional)
- Maintain patient records (optional)

Non-Functional Requirements

- High prediction accuracy
- Fast response time
- Secure handling of patient data
- User-friendly interface for ease of use

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